Neurosim: A Fast Simulator for Neuromorphic Robot Perception

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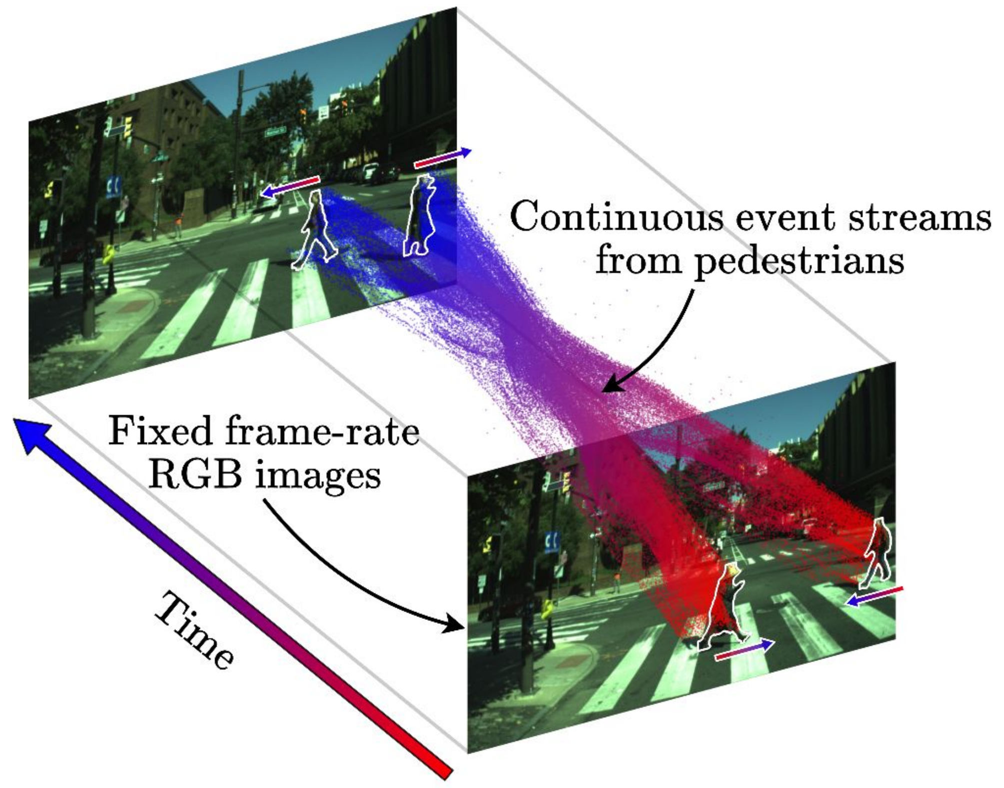
Event cameras provide high dynamic range, minimal temporal aliasing and consume very little power.



Captures fast scene dynamics

($< 100\,\mu s$ temporal resolution with ~ 1 ns precision)

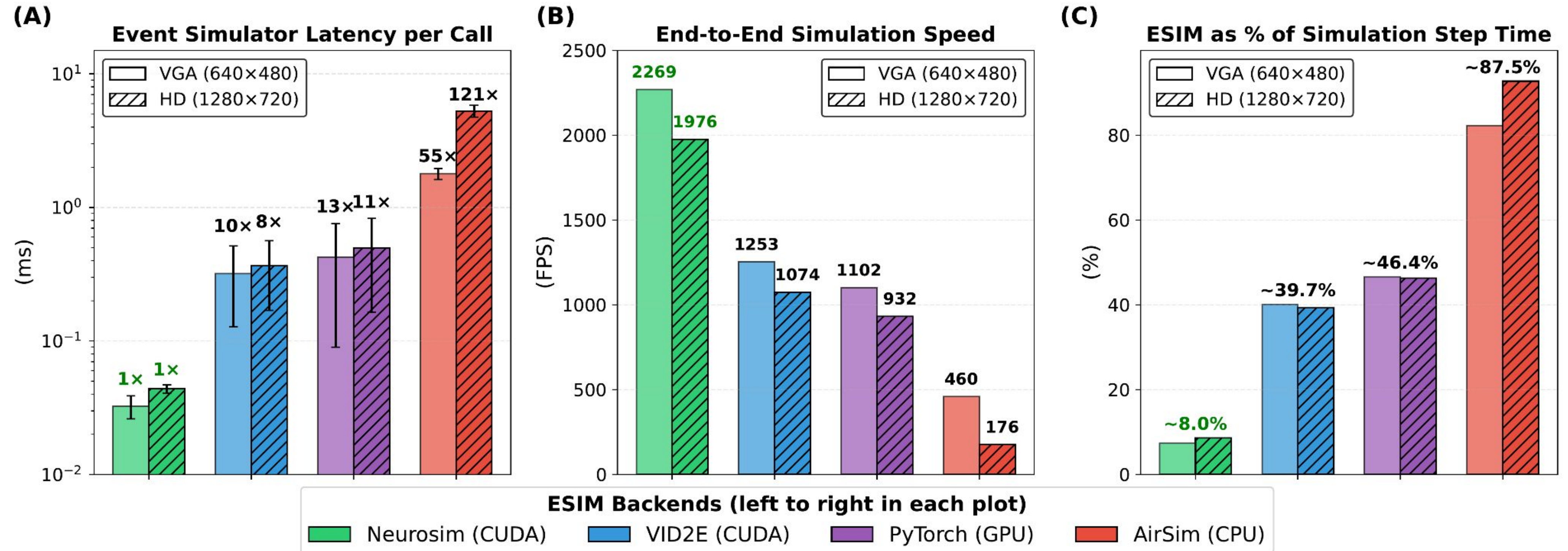
Handles extreme lighting conditions

($80 - 120$ dB dynamic range)

Is very power efficient

(< 10 mW peak power usage)

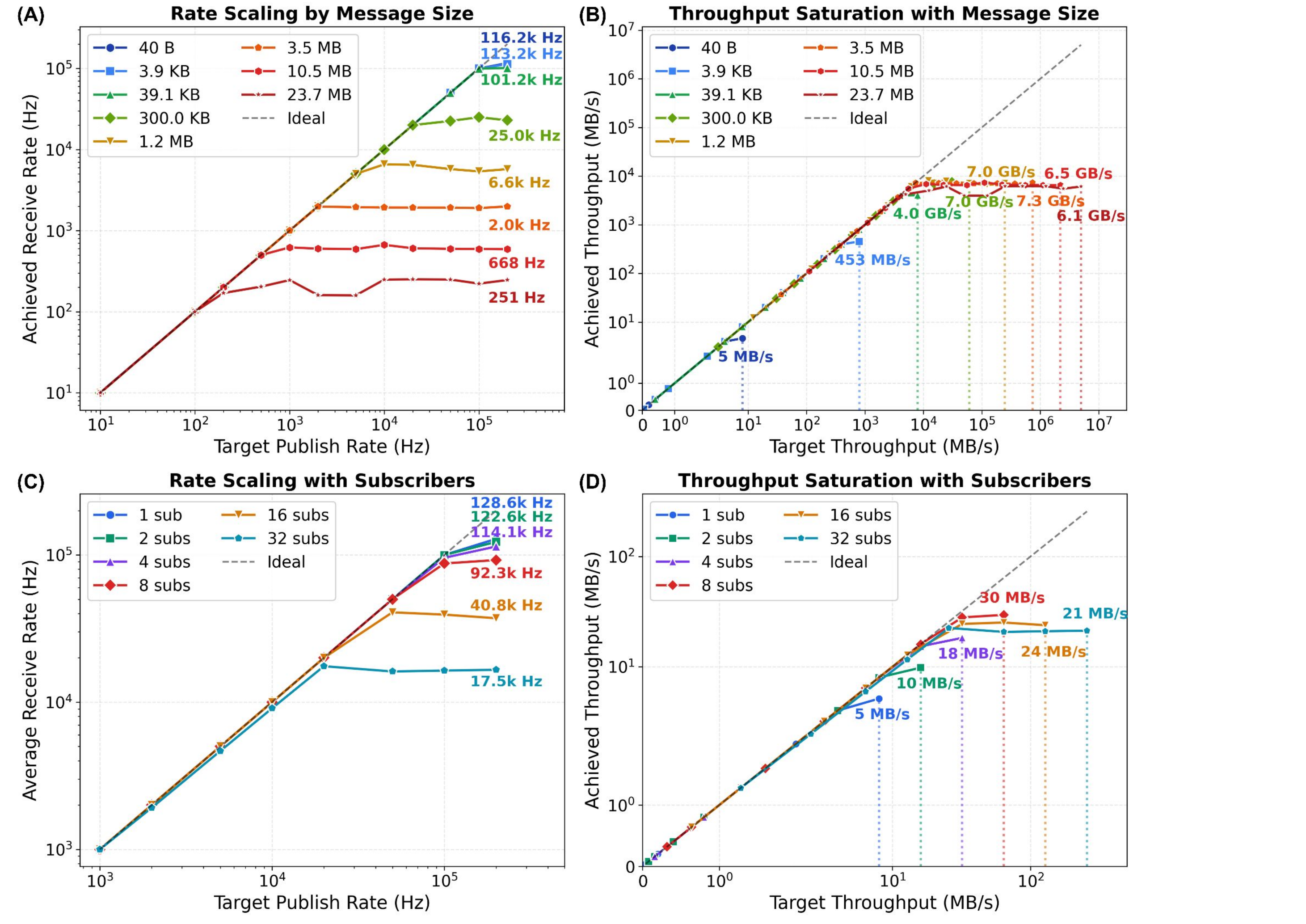
Neurosim simulates event cameras at multi-kilohertz rates



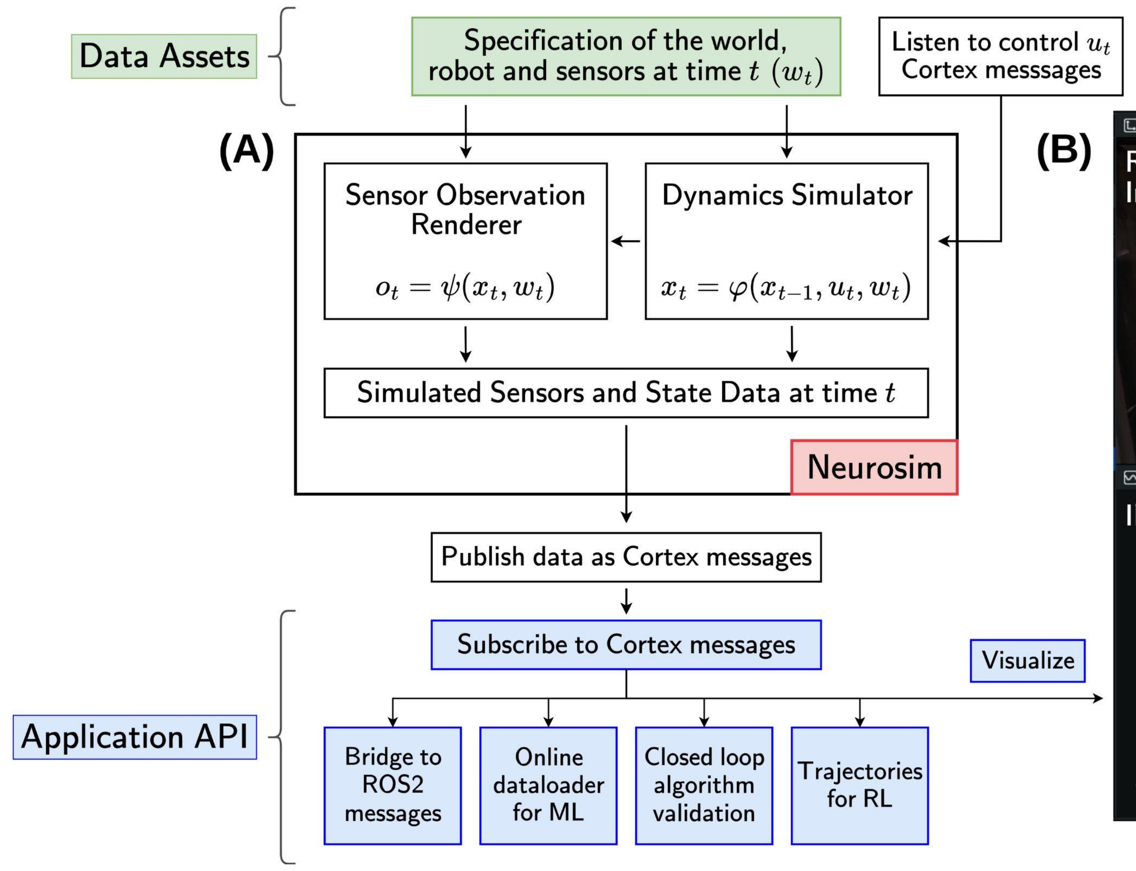
Warp-synchronous CUDA kernel design makes Neurosim's event camera simulation **10-13×** faster than existing GPU-based implementations. This prevents computational bottlenecks in Neurosim.

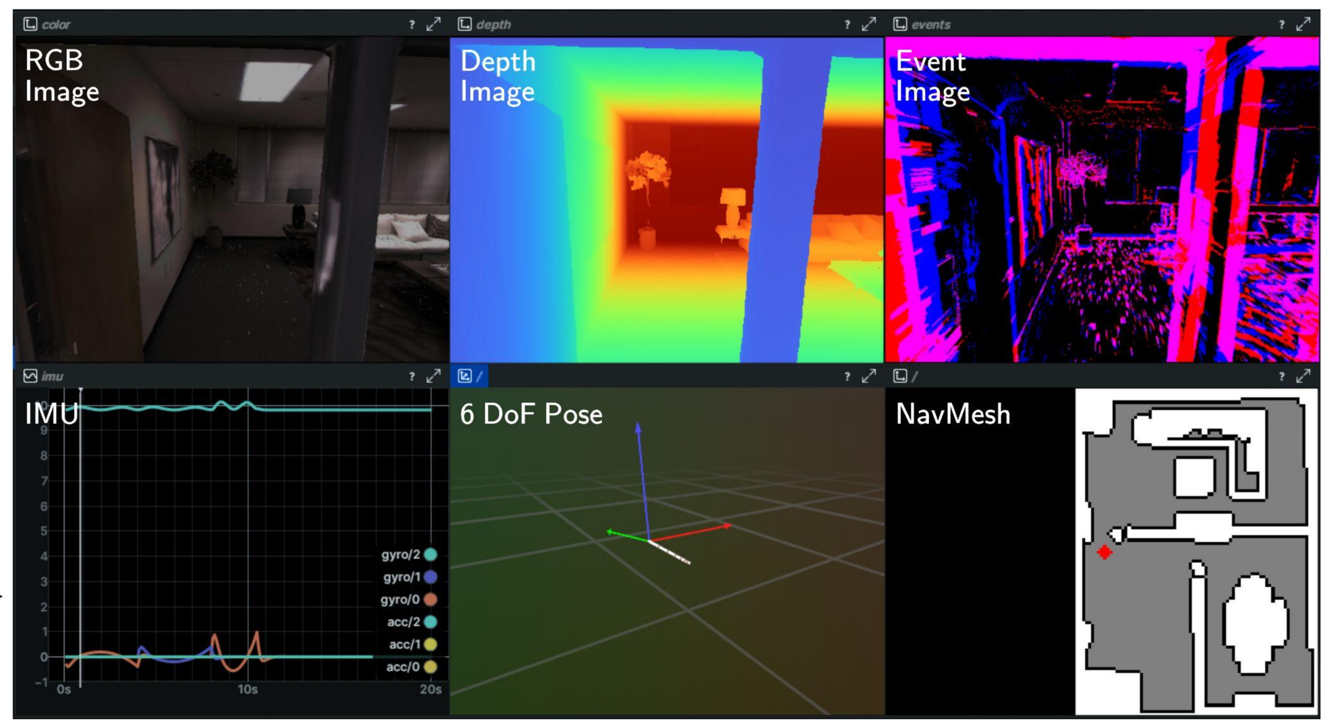
Cortex connects Neurosim to applications

Cortex is a high-performance ZeroMQ-based communication library for robotics and ML. It connects Neurosim to control loops, DL data loaders, hardware-in-the-loop, visualizers and more.



Neurosim is modular, high-performance (~2700 FPS), and easy to use for a variety of applications





Neurosim accurately simulates **RGB**, **depth**, **semantic** and **event**-based cameras; **inertial** and **optical flow** sensors; **edge** maps; **corner** detections; and **quadrotor dynamics** at **~2700 FPS** in indoor scenes on a single **RTX 4090** GPU. Additionally, it supports spawning/manipulating **dynamic obstacles** in the scene and **~250×** faster implementations of realistic event camera **noise models**.

Scan to try out Neurosim:
https://github.com/grasp-lyrl/neurosim

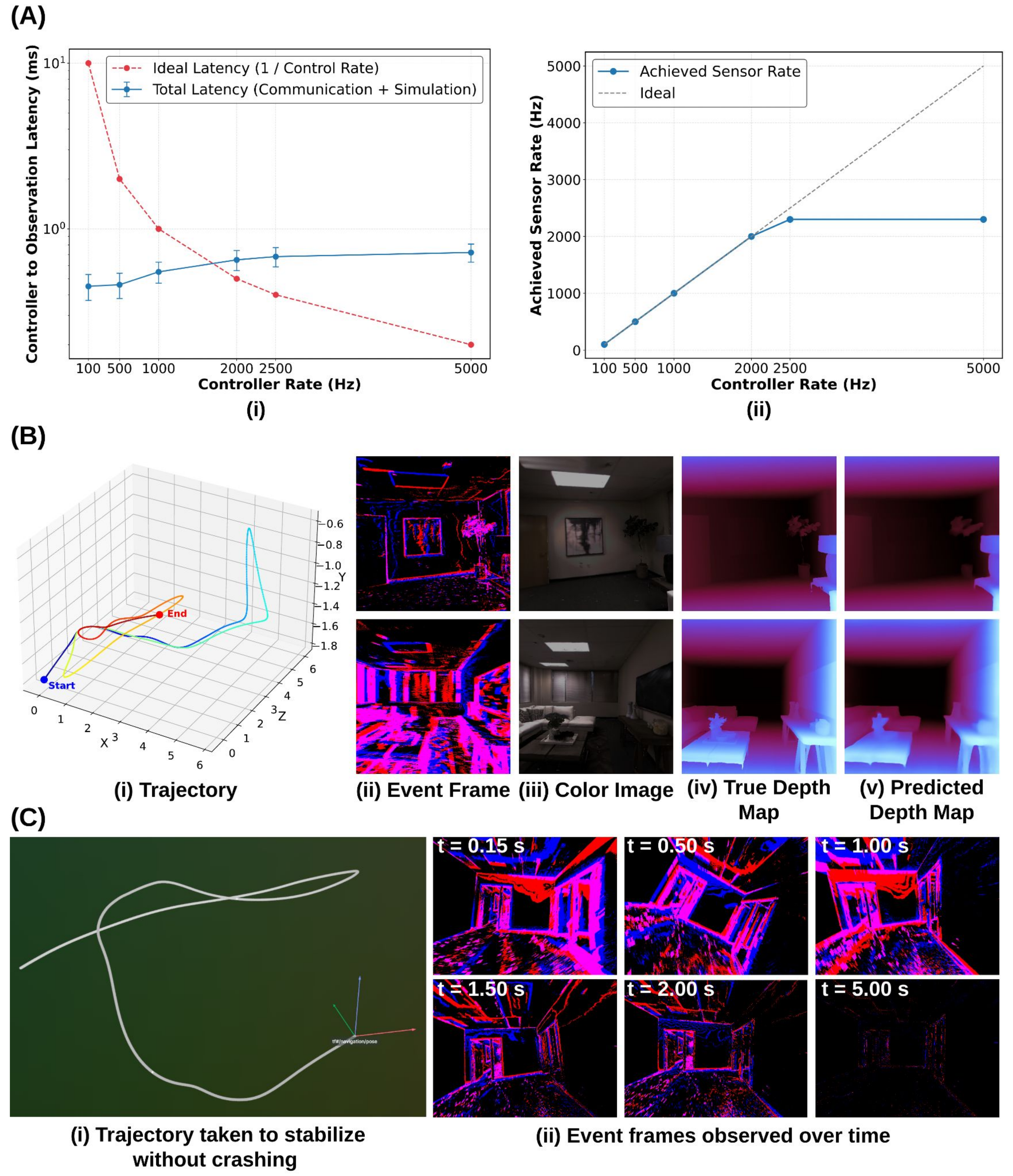
Scan to try out Cortex:
https://github.com/sudoRicheek/cortex

Neurosim can be used for closed-loop testing, online data generation for training and on-policy RL

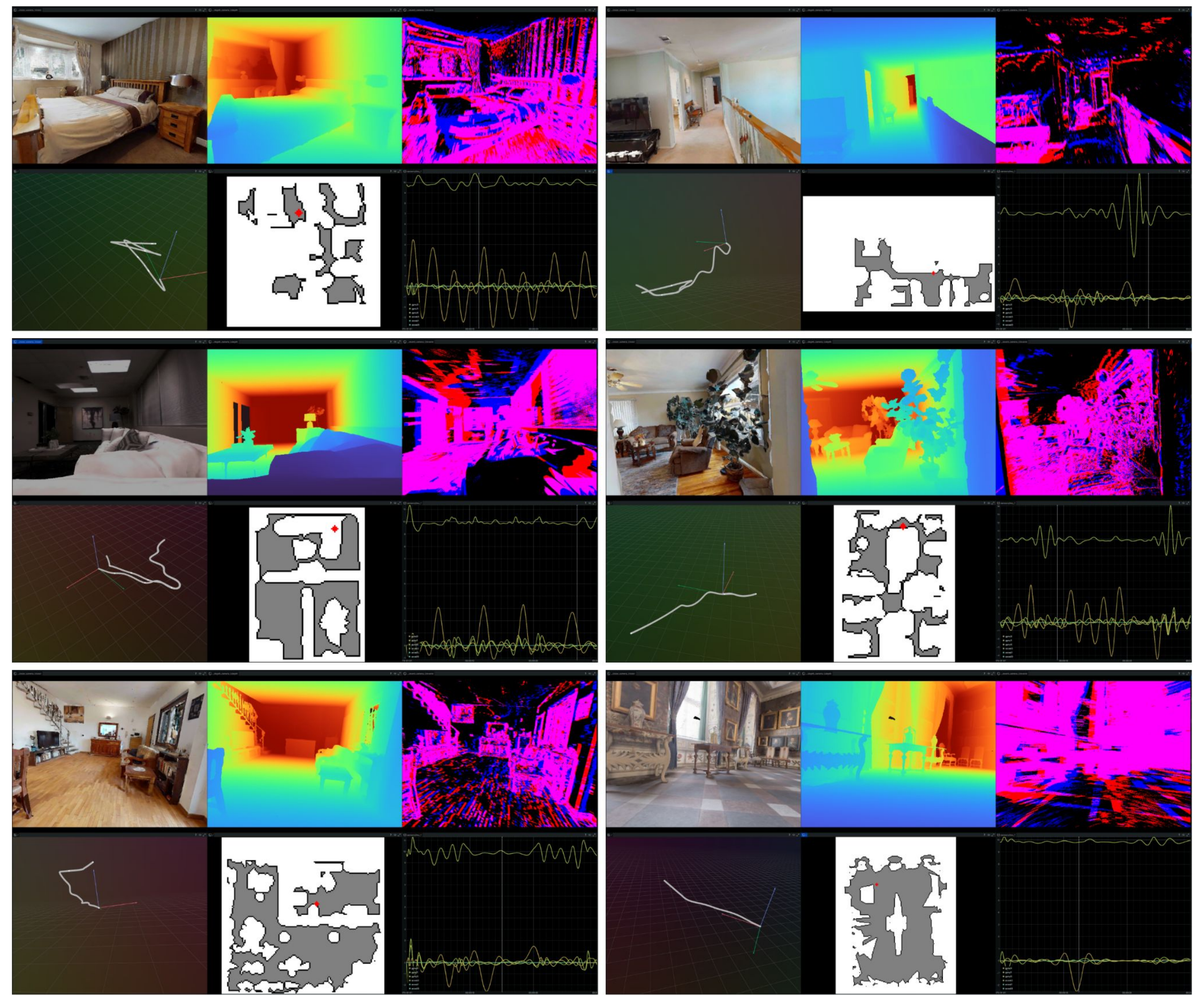
(A) With Cortex-based communication the controller respects sensor rates up to **2300 Hz** with **less than 0.7** ms of round-trip latency.

(B) Training this example event-based monocular depth model online saves **~450 GB per hour** of disk space that would otherwise store static simulated data.

(C) Example rollout of a policy learnt in Neurosim, completely online, to stabilize a thrown quadrotor in a cluttered indoor environment. The algorithm uses events and state estimates as inputs.



Examples of a quadrotor tracking random MinSnap trajectories in Neurosim



Trajectories across multiple floors, corridors, and interconnected rooms, with velocities often ≥ 4 rad/s (angular) and 2 m/s (linear).